We want to find an expression for the light-cone momentum and virtuality distribution, $\tilde{\rho}(\alpha, v)$ for deuterium. Segarra et al., Eq. 5, gives us:

$$\tilde{\rho}(\alpha, v) = \int dE d^3p S(E, \vec{p}) \frac{E + p_z}{E} \delta\left(\alpha - \frac{A(E + p_z)}{m_A}\right) \delta\left(v - \frac{E^2 - |p|^2 - m_N^2}{m_N^2}\right). \tag{1}$$

For deuterium, where the $(A-1)$ system is merely a nucleon, the spectral function is simply a momentum density, $n(p)$, with an energy-conserving δ-function:

$$S^d(E, p) = n(p)\delta\left(E - m_d + \sqrt{m_N^2 + p^2}\right). \tag{2}$$

For the derivation that follows, I will preserve the distinction between the proton and neutron mass. The nucleon is struck will be termed the "leading" nucleon, with subscript 'L,' while the spectator nucleon will receive subscript 's.' In this way, the spectral function for striking a nucleon of species 'L' can be written:

$$S_L^d(E, p) = n(p)\delta\left(E - m_d + \sqrt{m_s^2 + p^2}\right), \tag{3}$$

where the mass of spectator species enters into the δ-function.

Plugging this spectral function into equation 1 gives us:

$$\tilde{\rho}_L^d(\alpha, v) = \int dE d^3p n(p)\delta\left(E - m_d + \sqrt{m_s^2 + p^2}\right) \frac{E + p_z}{E} \delta\left(\alpha - \frac{2(E + p_z)}{m_d}\right) \delta\left(v - \frac{E^2 - |p|^2 - m_L^2}{m_L^2}\right). \tag{4}$$

It is simplest to attack this in cylindrical coordinates: $d^3p \to dp_z p_\perp dp_\perp d\phi$. We can eliminate the $p_\perp$ integral using the virtuality δ-function, from which we can see that $p_\perp = \sqrt{E^2 - p_z^2 + m_L^2(1 + v)}$. This yields:

$$\begin{aligned}\tilde{\rho}_L^d(\alpha, v) = \int dE dp_z d\phi p_\perp dp_\perp n(p)\delta\left(E - m_d + \sqrt{m_s^2 + p^2}\right) \frac{E + p_z}{E} \delta\left(\alpha - \frac{2(E + p_z)}{m_d}\right) \\ \cdot \frac{m_L^2}{2p_\perp} \delta\left(p_\perp - \sqrt{E^2 - p_z^2 + m_L^2(1 + v)}\right),\end{aligned} \tag{5}$$

which simplifies to

$$\tilde{\rho}_L^d(\alpha, v) = \frac{m_L^2}{2} \int dE dp_z d\phi n(p)\delta\left(E - m_d + \sqrt{m_s^2 + p^2}\right) \frac{E + p_z}{E} \delta\left(\alpha - \frac{2(E + p_z)}{m_d}\right). \tag{6}$$

The momentum magnitude, p, can be expressed as $p^2 = p_\perp^2 + p_z^2$, which transforms the penultimate delta function:

$$\tilde{\rho}_L^d(\alpha, v) = \frac{m_L^2}{2} \int dE dp_z d\phi n(p)\delta\left(E - m_d + \sqrt{m_s^2 + E^2 - m_L^2(1 + v)}\right) \frac{E + p_z}{E} \delta\left(\alpha - \frac{2(E + p_z)}{m_d}\right). \tag{7}$$

There is no sign of any ϕ dependence, so we may remove that integral,

$$\tilde{\rho}_L^d(\alpha, v) = \pi m_L^2 \int dE dp_z n(p)\delta\left(E - m_d + \sqrt{m_s^2 + E^2 - m_L^2(1 + v)}\right) \frac{E + p_z}{E} \delta\left(\alpha - \frac{2(E + p_z)}{m_d}\right), \tag{8}$$

and turn our attention to the δ-function specifying α. We can see that $p_z = \frac{m_d \alpha}{2} - E$, which allows us to write:

$$\tilde{\rho}_L^d(\alpha, v) = \pi m_L^2 \int dE dp_z n(p)\delta\left(E - m_d + \sqrt{m_s^2 + E^2 - m_L^2(1 + v)}\right) \frac{E + p_z}{E} \frac{m_d}{2} \delta\left(p_z - \frac{m_d \alpha}{2} + E\right), \tag{9}$$

perform the dp_z integral, and end up with:

$$\tilde{\rho}_L^d(\alpha, v) = \frac{\pi m_d^2 m_L^2}{4} \int \frac{dE}{E} n(p)\delta\left(E - m_d + \sqrt{m_s^2 + E^2 - m_L^2(1 + v)}\right). \tag{10}$$

To tackle the dE integral, we must transform the energy δ-function. Treating it as $\delta(f(E)$, where $f(E) = E - m_d + \sqrt{m_s^2 + E^2 - m_L^2(1+v)}$, we can take the derivative of f with respect to E:

$$f'(E) = 1 + \frac{E}{\sqrt{E^2 + m_s^2 - m_L^2(1+v)}}. \tag{11}$$

We can also invert f to get

$$E = \frac{m_d^2 + m_L^2(1+v) - m_s^2}{2m_d}. \tag{12}$$

This allows us to transform the energy δ-function to get:

$$\tilde{\rho}_L^d(\alpha, v) = \frac{\pi m_d^2 m_L^2}{4} \int \frac{dE}{E} n(p) \left| \frac{\sqrt{E^2 + m_s^2 - m_L^2(1+v)}}{E + \sqrt{E^2 + m_s^2 - m_L^2(1+v)}} \right| \delta\left(E - \frac{m_d^2 + m_L^2(1+v) - m_s^2}{2m_d} \right). \tag{13}$$

Performing the integral gives us:

$$\tilde{\rho}_L^d(\alpha, v) = \frac{\pi m_d^2 m_L^2}{4} \cdot \frac{n(p)}{E} \cdot \left| \frac{\sqrt{E^2 + m_s^2 - m_L^2(1+v)}}{E + \sqrt{E^2 + m_s^2 - m_L^2(1+v)}} \right|, \tag{14}$$

and we must now substitute in $\frac{m_d^2 + m_L^2(1+v) - m_s^2}{2m_d}$ for E.

Let's first attack the term in the square root:

$$E^2 = \frac{1}{4m_d^2} \left[m_d^4 + 2m_d^2 m_L^2(1+v) - 2m_d^2 m_s^2 + m_L^4(1+v)^2 - 2m_L^2 m_s^2(1+v) + m_s^4 \right] \tag{15}$$

$$m_s^2 - m_L^2(1+v) + E^2 = \frac{1}{4m_d^2} \left[m_d^4 - 2m_d^2 m_L^2(1+v) + 2m_d^2 m_s^2 + m_L^4(1+v)^2 - 2m_L^2 m_s^2(1+v) + m_s^4 \right] \tag{16}$$

$$m_s^2 - m_L^2(1+v) + E^2 = \left(\frac{1}{2m_d} \left[m_d^2 - m_L^2(1+v) + m_s^2 \right] \right)^2 \tag{17}$$

$$\sqrt{m_s^2 - m_L^2(1+v) + E^2} = \frac{\left| m_d^2 - m_L^2(1+v) + m_s^2 \right|}{2m_d} \tag{18}$$

Substituting this result into our expression gives us:

$$\tilde{\rho}_L^d(\alpha, v) = \frac{\pi m_d^2 m_L^2}{4} \cdot n(p) \cdot \frac{2m_d}{m_d^2 + m_L^2(1+v) - m_s^2} \cdot \left| \frac{m_d^2 - m_L^2(1+v) + m_s^2}{m_d^2 - m_L^2(1+v) + m_s^2 + m_d^2 + m_L^2(1+v) - m_s^2} \right| \tag{19}$$

$$= \frac{\pi m_d^2 m_L^2}{4} \cdot n(p) \cdot \frac{2m_d}{m_d^2 + m_L^2(1+v) - m_s^2} \cdot \left| \frac{m_d^2 - m_L^2(1+v) + m_s^2}{2m_d^2} \right| \tag{20}$$

$$= \frac{\pi m_d m_L^2}{4} \cdot n(p) \cdot \left| \frac{m_d^2 - m_L^2(1+v) + m_s^2}{m_d^2 + m_L^2(1+v) - m_s^2} \right| \tag{21}$$

Up until now, we have preserved the distinction between the proton mass, m_p, and the neutron mass, m_n. However, if ignore that distinction and assign both with the same nucleon mass, $m_N = m_s = m_L$, then our formula reduces to:

$$\tilde{\rho}^d(\alpha, v) = \frac{\pi m_d m_N^2}{4} \cdot n(p) \cdot \left| \frac{m_d^2 - v m_N^2}{m_d^2 + v m_N^2} \right| \tag{22}$$